

Assignment -Design Example

BCD Adder

Design a combinational circuit that adds two BCD digits and produces a valid BCD result.

Your design should:

- Detect when correction is required.
- Add 0110 when necessary.
- Produce the decimal carry.

Derive the correction-detection logic explicitly.

4-bit ALU Slice

Design a 4-bit ALU supporting the following operations:

Select	Operation
000	A+B
001	A-B
010	A+1
011	A-1
100	A AND B
101	A OR B
110	A XOR B
111	NOT A

Sequence-Dependent Combinational Mapping

A 4-bit input represents a hexadecimal digit. Design three outputs:

- $X = 1$ if the digit is a prime number.
- $Y = 1$ if the digit is divisible by 3.
- $Z = 1$ if the digit has an even number of 1s in its binary representation.

4-bit Binary to Two-Digit BCD Converter

Design a combinational circuit that accepts a 4-bit binary number 0–15 and outputs two BCD digits representing the decimal value.

Example :

$$1101_2 = 13_{10} \rightarrow 0001 \ 0011$$

